

$$\begin{aligned}\gamma(\xi) &= \alpha(\xi) - \frac{2}{l} \frac{d\omega(\xi)}{d\xi} \\ &= \frac{\xi}{2}(\xi-1)\alpha_1 + (1-\xi^2)\alpha_2 + \frac{\xi}{2}(\xi+1)\alpha_3 \\ &\quad - \frac{2}{l} \left(\left(\xi - \frac{1}{2}\right) \omega_1 - 2\xi \omega_2 + \left(\xi + \frac{1}{2}\right) \omega_3 \right)\end{aligned}$$

Strain,

$$\epsilon(\xi) = \begin{bmatrix} \kappa(\xi) \\ \gamma(\xi) \end{bmatrix} = \begin{bmatrix} 0 & \frac{2}{l} N_1' & 0 & \frac{2}{l} N_2' & 0 & \frac{2}{l} N_3' \\ -\frac{2}{l} N_1' & N_1 & -\frac{2}{l} N_2' & N_2 & -\frac{2}{l} N_3' & N_3 \end{bmatrix} \underline{u} = \underline{B} \underline{u}$$

$$\therefore \begin{bmatrix} 0 & \frac{2\xi-1}{l} & 0 & \frac{-4\xi}{l} & 0 & \frac{2\xi+1}{l} \\ \frac{1-2\xi}{l} & \frac{\xi}{2}(\xi-1) & \frac{4\xi}{l} & 1-\xi^2 & -\frac{2\xi+1}{l} & \frac{\xi}{2}(\xi+1) \end{bmatrix} \underline{u} \\ = \underline{B} \underline{u}$$

(iii) Constitutive

$$\underline{\sigma} = \begin{bmatrix} EI & 0 \\ 0 & GA_s \end{bmatrix}$$

$$\sigma(\xi) = \underline{\sigma} \underline{B} \underline{u}$$

(iv) Element stiffness Matrix,

$$K_e = \int_{-1}^1 \underline{B}^T \underline{\sigma} \underline{B} d\xi J = \frac{l}{2} \int_{-1}^1 \underline{B}^T \underline{\sigma} \underline{B} d\xi$$

$$\underline{\underline{K_B}} = \begin{bmatrix} EI & 0 \\ 0 & GA_s \end{bmatrix} \begin{bmatrix} 0 & \frac{2\xi-1}{l} & 0 & -\frac{4\xi}{l} & 0 & \frac{2\xi+1}{l} \\ \frac{1-2\xi}{l} & \frac{\xi(\xi-1)}{2} & \frac{4\xi}{l} & 1-\xi^2 & -\frac{2\xi+1}{l} & \frac{\xi(\xi+1)}{2} \end{bmatrix}$$

$$\underline{\underline{B^T K_B}} = \begin{bmatrix} 0 & \frac{1-2\xi}{l} \\ \frac{2\xi-1}{l} & \frac{\xi(\xi-1)}{2} \\ 0 & \frac{4\xi}{l} \\ -\frac{4\xi}{l} & 1-\xi^2 \\ 0 & -\frac{2\xi+1}{l} \\ \frac{2\xi+1}{l} & \frac{\xi(\xi+1)}{2} \end{bmatrix} \begin{bmatrix} 0 & \frac{EI(2\xi-1)}{l} & 0 & -\frac{EI(4\xi)}{l} & 0 & \frac{EI(2\xi+1)}{l} \\ \frac{GA_s(1-2\xi)}{l} & \frac{GA_s \xi(\xi-1)}{2} & \frac{GA_s 4\xi}{l} & GA_s(1-\xi^2) & -\frac{GA_s(2\xi+1)}{l} & \frac{GA_s \xi(\xi+1)}{2} \end{bmatrix}$$

This yields the same stiffness matrix,

$$K_e = \begin{bmatrix} \frac{7GA_s}{3l} & \frac{GA_s}{2} & -\frac{8GA_s}{3l} & \frac{2GA_s}{3} & \frac{GA_s}{3l} & -\frac{GA_s}{6} \\ \frac{7EI}{3l} + \frac{2GA_s l}{15} & \frac{-2GA_s}{3} & \frac{-8EI}{3l} + \frac{GA_s l}{15} & \frac{GA_s}{6} & \frac{EI}{3l} - \frac{GA_s l}{30} \\ \frac{16GA_s}{3l} & 0 & \frac{-8GA_s}{3l} & \frac{2GA_s}{3} \\ \frac{16EI}{3l} + \frac{8GA_s l}{15} & -\frac{2GA_s}{3} & \frac{-8EI}{3l} + \frac{GA_s l}{15} \\ \frac{7GA_s}{3l} & -\frac{GA_s}{2} \\ \frac{7EI}{3l} + \frac{2GA_s l}{15} \end{bmatrix}$$